II. Daron Acemoglu, Simon Johnson, and James Robinson

"The Colonial Origins of Comparative Development: An Empirical Investigation"

AJR's Thesis

- Settler mortality affected colonialization strategy, which affected institutions.

- These institutional differences have persisted.

- Engerman and Sokoloff advance similar ideas; but they focus on conduciveness to slave agriculture rather than the disease environment.

AJR's Basic Empirical Strategy

- In a sample of former colonies, regress income per capital today on institutions today, instrumenting with settler mortality.
- "This identification strategy will be valid as long as … mortality rates of settlers between the seventeenth and nineteenth centuries have no effect on income today other than through their influence on institutional development" (AJR, p. 1383).
- No!! The key issue is whether settler mortality is correlated with determinants of income today other than institutions.

AJR’s Qualitative Evidence

- Mortality influenced settlement patterns.
- Colonizers adopted very different strategies in different places: “settler colonies” vs. “extractive states.”
- Institutions had considerable persistence.
- Evaluation?

OLS

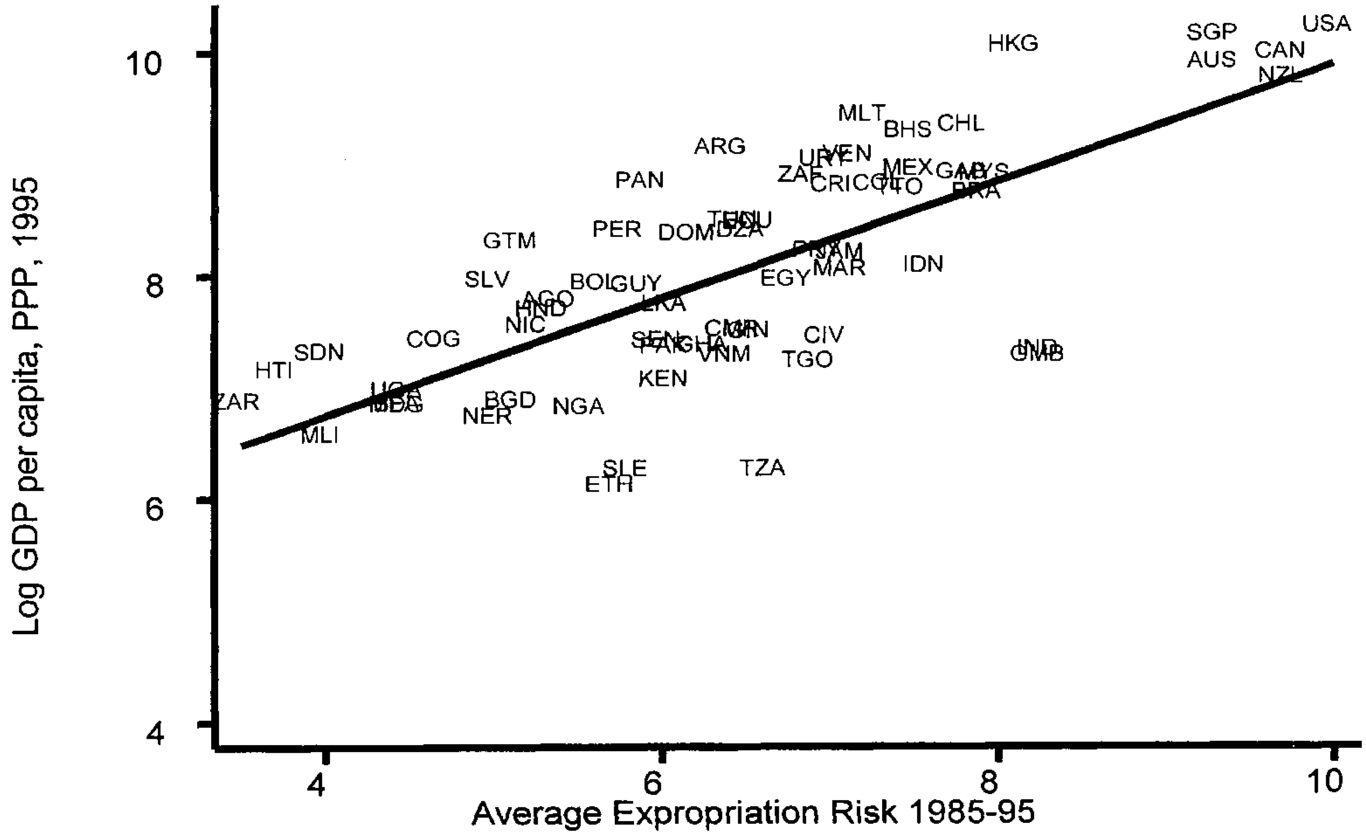

FIGURE 2. OLS RELATIONSHIP BETWEEN EXPROPRIATION RISK AND INCOME

From: AJR, “The Colonial Origins of Comparative Development”

TABLE 2—OLS REGRESSIONS

	Whole world (1)	Base sample (2)	Whole world (3)	Whole world (4)	Base sample (5)	Base sample (6)	Whole world (7)	Base sample (8)
	Dependent variable is log GDP per capita in 1995						Dependent variable is log output per worker in 1988	
Average protection against expropriation risk, 1985–1995	0.54 (0.04)	0.52 (0.06)	0.47 (0.06)	0.43 (0.05)	0.47 (0.06)	0.41 (0.06)	0.45 (0.04)	0.46 (0.06)
Latitude			0.89 (0.49)	0.37 (0.51)	1.60 (0.70)	0.92 (0.63)		
Asia dummy				−0.62 (0.19)		−0.60 (0.23)		
Africa dummy				−1.00 (0.15)		−0.90 (0.17)		
"Other" continent dummy				−0.25 (0.20)		−0.04 (0.32)		
R^2	0.62	0.54	0.63	0.73	0.56	0.69	0.55	0.49
Number of observations	110	64	110	110	64	64	108	61

Notes: Dependent variable: columns (1)–(6), log GDP per capita (PPP basis) in 1995, current prices (from the World Bank's World Development Indicators 1999); columns (7)–(8), log output per worker in 1988 from Hall and Jones (1999). Average protection against expropriation risk is measured on a scale from 0 to 10, where a higher score means more protection against expropriation, averaged over 1985 to 1995, from Political Risk Services. Standard errors are in parentheses. In regressions with continent dummies, the dummy for America is omitted. See Appendix Table A1 for more detailed variable definitions and sources. Of the countries in our base sample, Hall and Jones do not report output per worker in the Bahamas, Ethiopia, and Vietnam.

From: AJR, "The Colonial Origins of Comparative Development"

Data on Potential Settler Mortality

- Mainly death rates of soldiers (not from battle).
- For Latin America, mainly based on death rates of bishops, adjusted to reflect higher death rates of soldiers.
- Deaths were largely from disease, especially malaria and yellow fever.
- AJR argue that the diseases had much smaller effects on local populations.

IV – First Stage

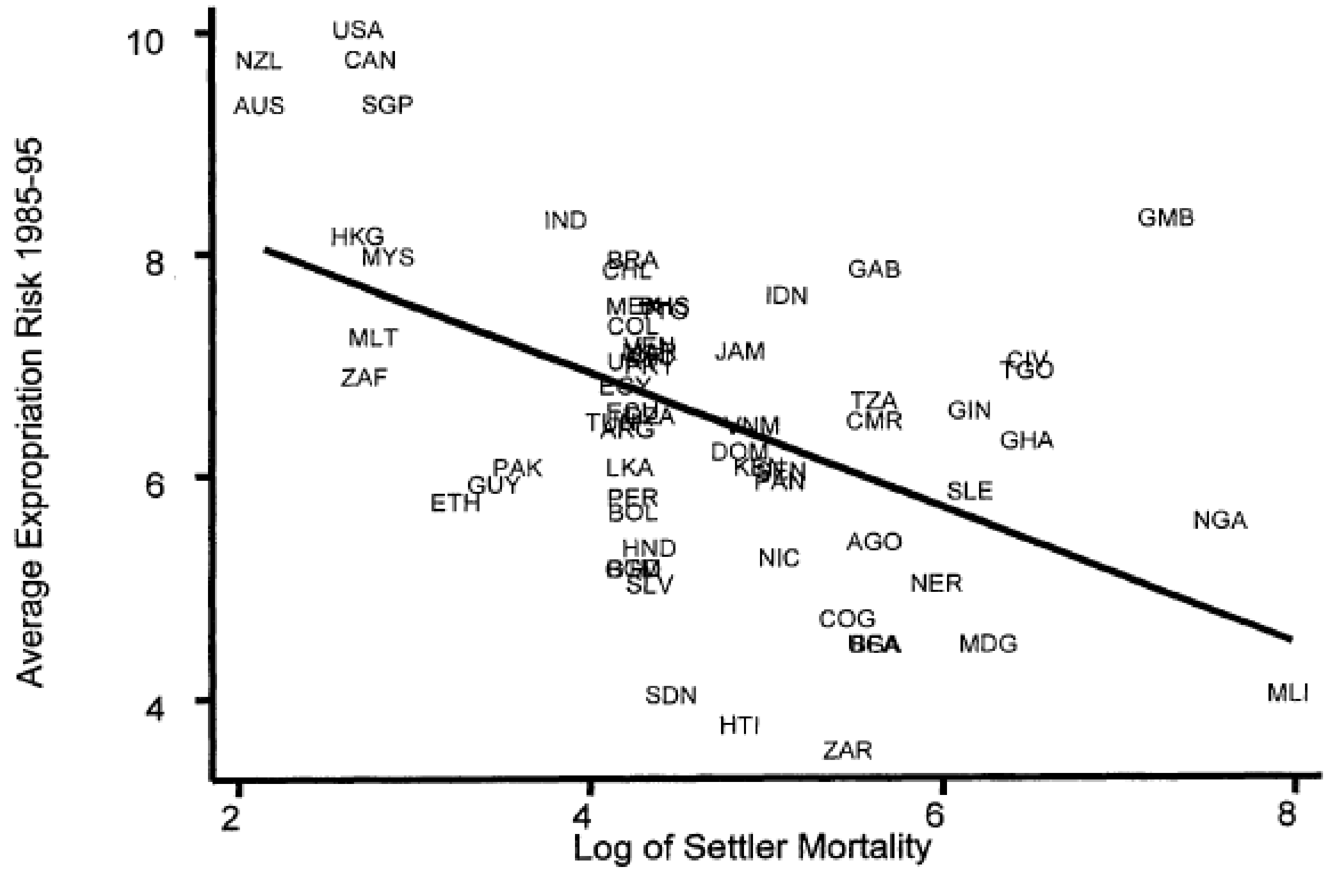

FIGURE 3. FIRST-STAGE RELATIONSHIP BETWEEN SETTLER MORTALITY AND EXPROPRIATION RISK

From: AJR, “The Colonial Origins of Comparative Development”

TABLE 4—IV REGRESSIONS OF LOG GDP PER CAPITA

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk 1985–1995	0.94	1.00	1.28	1.21	0.58	0.58	0.98	1.10	0.98
	(0.16)	(0.22)	(0.36)	(0.35)	(0.10)	(0.12)	(0.30)	(0.46)	(0.17)
Latitude		−0.65		0.94		0.04		−1.20	
		(1.34)		(1.46)		(0.84)		(1.8)	
Asia dummy							−0.92	−1.10	
							(0.40)	(0.52)	
Africa dummy							−0.46	−0.44	
							(0.36)	(0.42)	
"Other" continent dummy							−0.94	−0.99	
							(0.85)	(1.0)	
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.61	−0.51	−0.39	−0.39	−1.20	−1.10	−0.43	−0.34	−0.63
	(0.13)	(0.14)	(0.13)	(0.14)	(0.22)	(0.24)	(0.17)	(0.18)	(0.13)
Latitude		2.00		−0.11		0.99		2.00	
		(1.34)		(1.50)		(1.43)		(1.40)	
Asia dummy							0.33	0.47	
							(0.49)	(0.50)	
Africa dummy							−0.27	−0.26	
							(0.41)	(0.41)	
"Other" continent dummy							1.24	1.1	
							(0.84)	(0.84)	
R^2	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.28
Panel C: Ordinary Least Squares									
Average protection against expropriation risk 1985–1995	0.52	0.47	0.49	0.47	0.48	0.47	0.42	0.40	0.46
	(0.06)	(0.06)	(0.08)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)
Number of observations	64	64	60	60	37	37	64	64	61

From: AJR, "The Colonial Origins of Comparative Development"

Discussion

- Latitude and Africa dummy (vs. Americas) are insignificant (!).
- OLS vs. IV: Can measurement error – broadly defined – plausibly explain why the IV estimates are so much larger?
- If the measurement error is classical, $\hat{b}_{OLS} = \frac{V_X}{V_X + V_u} b_{TRUE}$, where V_X and V_u are the variances of the "true" X and of the measurement error.
- Implied economic importance from the IV estimate?

Candidates for Omitted Variables Correlated with the Instrument

- Identity of the colonizer.
- Legal origins.
- Religion.
- Weather.
- Suitability for agriculture.
- Modern disease environment.
- Effects of the slave trade operating through culture rather than institutions.
- Human capital accumulation.
- More?

TABLE 6—ROBUSTNESS CHECKS FOR IV REGRESSIONS OF LOG GDP PER CAPITA

	Base sample (1)	Base sample (2)	Base sample (3)	Base sample (4)	Base sample (5)	Base sample (6)	Base sample (7)	Base sample (8)	Base sample (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk, 1985–1995	0.84	0.83	0.96	0.99	1.10	1.30	0.74	0.79	0.71
	(0.19)	(0.21)	(0.28)	(0.30)	(0.33)	(0.51)	(0.13)	(0.17)	(0.20)
Latitude		0.07		−0.67		−1.30		−0.89	−2.5
		(1.60)		(1.30)		(2.30)		(1.00)	(1.60)
p-value for temperature variables	[0.96]	[0.97]							[0.77]
p-value for humidity variables	[0.54]	[0.54]							[0.62]
Percent of European descent in 1975			−0.08	0.03					0.3
			(0.82)	(0.84)					(0.7)
p-value for soil quality					[0.79]	[0.85]			[0.46]
p-value for natural resources					[0.82]	[0.87]			[0.82]
Dummy for being landlocked					0.64	0.79			0.75
					(0.63)	(0.83)			(0.47)
Ethnolinguistic fragmentation							−1.00	−1.10	−1.60
							(0.32)	(0.34)	(0.47)
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.64	−0.59	−0.41	−0.4	−0.44	−0.34	−0.64	−0.56	−0.59
	(0.17)	(0.17)	(0.14)	(0.15)	(0.16)	(0.17)	(0.15)	(0.15)	(0.21)
Latitude		2.70		0.48		2.20		2.30	4.20
		(2.00)		(1.50)		(1.50)		(1.40)	(2.60)
R^2	0.39	0.41	0.34	0.34	0.41	0.43	0.27	0.30	0.59

From: AJR, "The Colonial Origins of Comparative Development"

TABLE 7—GEOGRAPHY AND HEALTH VARIABLES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Instrumenting only for average protection against expropriation risk						Instrumenting for all right-hand-side variables			Yellow fever instrument for average protection against expropriation risk	
Panel A: Two-Stage Least Squares											
Average protection against expropriation risk, 1985–1995	0.69	0.72	0.63	0.68	0.55	0.56	0.69	0.74	0.68	0.91	0.90
	(0.25)	(0.30)	(0.28)	(0.34)	(0.24)	(0.31)	(0.26)	(0.24)	(0.23)	(0.24)	(0.32)
Latitude		−0.57		−0.53		−0.1					
		(1.04)		(0.97)		(0.95)					
Malaria in 1994	−0.57	−0.60					−0.62				
	(0.47)	(0.47)					(0.68)				
Life expectancy			0.03	0.03				0.02			
			(0.02)	(0.02)				(0.02)			
Infant mortality					−0.01	−0.01			−0.01		
					(0.005)	(0.006)			(0.01)		
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995											
Log European settler mortality	−0.42	−0.38	−0.34	−0.30	−0.36	−0.29	−0.41	−0.40	−0.40		
	(0.19)	(0.19)	(0.17)	(0.18)	(0.18)	(0.19)	(0.17)	(0.17)	(0.17)		
Latitude		1.70		1.10		1.60	−0.81	−0.84	−0.84		
		(1.40)		(1.40)		(1.40)	(1.80)	(1.80)	(1.80)		
Malaria in 1994	−0.79	−0.65									
	(0.54)	(0.55)									
Life expectancy			0.05	0.04							
			(0.02)	(0.02)							
Infant mortality					−0.01	−0.01					
					(0.01)	(0.01)					
Mean temperature							−0.12	−0.12	−0.12		
							(0.05)	(0.05)	(0.05)		
Distance from coast							0.57	0.55	0.55		
							(0.51)	(0.52)	(0.52)		
Yellow fever dummy										−1.10	−0.81
										(0.41)	(0.38)
R^2	0.3	0.31	0.34	0.35	0.32	0.34	0.37	0.36	0.36	0.10	0.32

From: AJR, “The Colonial Origins of Comparative Development”

Other Issues

- Concerns abut the data: Albouy (2012 and others) vs. AJR (2012 and others).

- Are the intermediate steps (e.g., institutions in 1900 and at time of independence) strong enough?

TABLE 3—DETERMINANTS OF INSTITUTIONS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A	Dependent Variable Is Average Protection Against Expropriation Risk in 1985–1995									
Constraint on executive in 1900	0.32	0.26								
	(0.08)	(0.09)								
Democracy in 1900			0.24	0.21						
			(0.06)	(0.07)						
Constraint on executive in first year of independence					0.25	0.22				
					(0.08)	(0.08)				
European settlements in 1900							3.20	3.00		
							(0.61)	(0.78)		
Log European settler mortality									−0.61	−0.51
									(0.13)	(0.14)
Latitude		2.20		1.60		2.70		0.58		2.00
		(1.40)		(1.50)		(1.40)		(1.51)		(1.34)
R^2	0.2	0.23	0.24	0.25	0.19	0.24	0.3	0.3	0.27	0.3
Number of observations	63	63	62	62	63	63	66	66	64	64
Panel B	Dependent Variable Is Constraint on Executive in 1900				Dependent Variable Is Democracy in 1900				Dependent Variable Is European Settlements in 1900	
European settlements in 1900	5.50	5.40			8.60	8.10				
	(0.73)	(0.93)			(0.90)	(1.20)				
Log European settler mortality			−0.82	−0.65			−1.22	−0.88	−0.11	−0.07
			(0.17)	(0.18)			(0.24)	(0.25)	(0.02)	(0.02)
Latitude		0.33		3.60		1.60		7.60		0.87
		(1.80)		(1.70)		(2.30)		(2.40)		(0.19)
R^2	0.46	0.46	0.25	0.29	0.57	0.57	0.28	0.37	0.31	0.47
Number of observations	70	70	75	75	67	67	68	68	73	73

From: AJR, “The Colonial Origins of Comparative Development”

Conclusion